Indian TAL BRABO robot ready for Europe

After successfully demonstrating India's first made in India Robot called BRABO at Make in India Week held last year, TAL Manufacturing, a wholly-owned subsidiary of Tata Motors, has received CE certification for BRABO, for the commercial sale of the robot in European markets.

An innovative concept developed from an industrial user perspective to automate micro, small and medium enterprises, TAL BRABO now complies with essential requirements relevant to European health, safety and environmental legislation, permitting export of TAL BRABO robot to customers in Europe.

Designed indigenously, BRABO can be used to create efficiencies right from raw material handling, until packaging of the finished product. It can be programmed to operate 24/7 in all situations, for continuous production, with a high degree of flexibility, and even perform complex functions cost-effectively.

BRABO can be used for varied applications for tasks like pick and placement of materials, assembly of parts, machine and press tending, as a sealing application, camera and vision based jobs. The robot can also complement human workforce by taking over dull, dangerous and monotonous jobs.

With BRABO, TAL Manufacturing will focus on automotive, electronics, logistics, food, packaging, pharmaceutical and other industries. The current list of BRABO customers includes TATA Motors, Mahindra & Mahindra, Diebold, CPG Industries, Hydromatik, SGK Industries, RTA Spa Italy, BITS Dubai Campus, Suparna Plastics, Micromax Systems, Twin Engineers, AM Ecosystems and Kaziranga University.

BRABO was designed and styled in-house at TAL Manufacturing and Tata Elxsi, respectively, while manufacturing of some critical parts was done at Tata AutoComp (Image courtesy: www.tal.co.in)

Robot skin that is more sensitive than a human hand

Engineers at University of Glasgow, Scotland, who had previously developed an electronic skin covering for prosthetic hands made from graphene, have found a way to use some of graphene's remarkable physical properties to use energy from the Sun to power the skin.

Graphene is a highly-flexible form of graphite which, despite being just a single atom thick, is stronger than steel, electrically-conductive and transparent. It is graphene's optical transparency—which allows around 98 per cent of the light that strikes its surface to pass directly through it—that makes it ideal for gathering energy from the Sun to generate power.

The new skin requires only 20 nanowatts of power per square centimetre, which is easily met even by the poorest-quality photovoltaic cells currently available in the market. Although, currently, the energy generated by the skin's photovoltaic cells cannot be stored, the team is already looking into ways to divert unused energy into batteries, allowing the energy to be used


Prof. Ravinder Dahiya showing off a prosthetic hand covered with his new, highly-sensitive graphene skin (Image courtesy: www.forbes.com)